

Eye-Tracking Driven Personalized 360° Panorama Content Generation

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ABSTRACT

Personalized 360° content generation faces significant challenges due to the lack of adaptation to user preferences during interaction. Traditional methods often rely on static content or post-session analysis, which limits dynamic engagement. In this paper, we propose a novel approach (*i.e.*, DynamicEye360) that integrates eye-tracking with dynamic content generation to create personalized 360° experiences. By analyzing gaze trajectories and user preferences, our approach adjusts content based on the user's visual focus, leading to a more immersive and engaging experience. Experimental results show that our approach significantly enhances content relevance and user engagement compared to existing methods, demonstrating the potential of gaze-driven personalized content in virtual environments.

Index Terms: 360° Panoramic Content, Personalized Content Generation, User Interaction.

1 INTRODUCTION

360° content has become a key element in virtual reality (VR) and immersive environments, providing users with an engaging experience by enabling exploration from any viewpoint. However, current 360° content generation methods are limited in their ability to adapt to individual user preferences. Traditional approaches often rely on pre-generated content or post-session analysis, which do not consider user interactions during the experience. This lack of dynamic adaptation reduces both immersion and personalization.

Advances in eye-tracking technology offer an opportunity to overcome this limitation by capturing users' gaze and attention. Eye-tracking data offers valuable insights into user preferences, enabling more accurate content tailoring. However, integrating gaze tracking into content generation seamlessly remains challenging, especially in 360° environments where content must evolve as users interact with the scene.

In this paper, as shown in Fig. 1, we propose a novel approach that combines gaze tracking with dynamic 360° content generation. By analyzing gaze data to understand user preferences, our DynamicEye360 adjusts content based on the user's visual focus, enhancing both personalization and immersion. Experiments demonstrate significant improvements in user engagement and content relevance compared to existing methods. Our approach represents progress toward truly adaptive 360° content that evolves to meet the needs and preferences of individual users.

2 RELATED WORK

Recent research in 360° content generation has focused on enhancing immersion and user experience, primarily through static content or predefined interactions. Traditional methods [3, 1] often rely on user input, such as clicks or post-session analysis, which limits dynamic personalization. Some studies have explored gaze-based interaction in VR and AR, using eye-tracking for adaptive content,

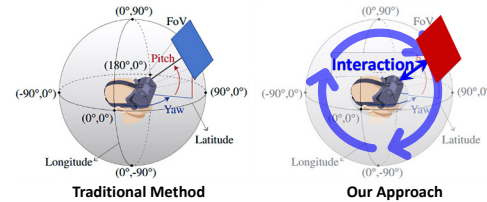


Figure 1: Comparison between traditional method and our approach. In traditional method, users can only view the content played by the HMD, while in our approach, users interact with the content as the HMD collects gaze data and updates the scene, enabling continuous iterative interaction.

but these approaches generally focus on pre-defined scenes or static environments.

In contrast, our approach integrates gaze tracking with dynamic content generation, enabling personalized 360° experiences that adapt to users' gaze patterns and preferences. This method improves on existing systems by continuously adjusting content based on human gaze data, providing a more immersive experience.

3 METHODOLOGY

3.1 DynamicEye360 Overview

In this paper, we propose a method for personalized 360° content generation that integrates gaze data collection, user preference analysis, and dynamic content adjustment. Below, we describe the components of our method in detail, starting with the gaze data collection process, followed by preference analysis, and finally, the dynamic content generation mechanism.

3.2 Gaze Data Collection and Processing

Traditional gaze data collection methods [4, 2] are often disconnected from the content generation process, where content is displayed first, followed by gaze data collection and separate preference analysis. This traditional approach prevents adaptive changes to the content based on user interaction, limiting personalization and immersion.

To address this limitation, our method combines gaze data collection with content generation, allowing for continuous adaptation of the content to the user's visual focus. Gaze data, captured via eye-tracking using the Head-Mounted Display (HMD), is used to identify regions of interest in the 360° scene. This data is then processed to build a user preference model that reflects the user's visual engagement with the content. The model outputs a personalized preference profile, which is then used to dynamically adjust the content, prioritizing the areas the user finds most engaging.

As shown in Fig. 2-A, the DynamicEye360 captures gaze data from the HMD and transmits it to the server. Then, the data is analyzed by a large language model to assess user preferences, and the results are used to adjust and generate personalized 360° content. The content is then transmitted back to the HMD for display, forming a closed-loop DynamicEye360 that ensures the content continuously aligns with the user's attention. This approach allows for a highly adaptive, immersive experience that evolves based on the user's focus and preferences.

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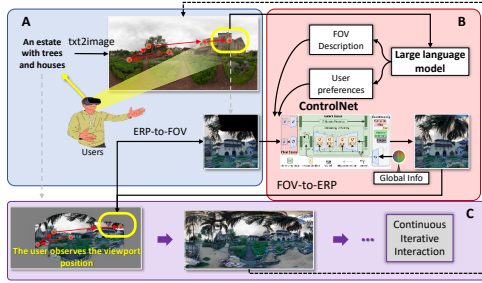


Figure 2: Overview of the Interactive 360° Content Generation Process. This diagram illustrates the workflow of our method, where gaze data is continuously collected by the HMD, analyzed to determine user preferences, and used to dynamically adjust the content. The process iterates, ensuring that the displayed scene adapts according to the user's visual focus, providing an interactive and personalized experience.

3.3 User Preference Analysis and Modeling

Traditional 360° content generation methods [3, 1] typically rely on explicit feedback or predefined rules, resulting in static and disconnected approaches that fail to adapt to users' attention. These methods overlook gaze behavior, thus limiting personalization and reducing user engagement. In contrast, gaze data offers insights into user preferences by revealing where users focus and how they explore content. By leveraging this information, content can be dynamically tailored to the user's visual attention, creating a more responsive and engaging experience.

To address the limitations of existing methods, as shown in Fig. 2-B, we propose a novel solution that integrates gaze data analysis using a large language model (LLM) [5]. When the DynamicEye360 receives gaze and Field of View (FOV) data from the HMD, the LLM is used to analyze user preferences. The LLM model takes both the displayed 360° content and real-time gaze trajectories as input, enabling a dynamic assessment of the user's attention. The output is a personalized preference profile that reflects the user's focus areas and engagement levels.

Our approach allows continuous updates to the user's preferences, adapting the content as their attention shifts. By combining gaze behavior with contextual content, we ensure the 360° scene evolves in real time, providing an immersive and personalized experience that aligns closely with the user's individual interests.

3.4 Dynamic 360° Content Generation

Once user preferences and FOV content descriptions are obtained, the next step is to incorporate them into the generation of personalized 360° content. Rather than creating a static scene, our approach dynamically adjusts the content based on user feedback, generating a new FOV at each step to ensure the scene evolves in response to the user's gaze and preferences.

As illustrated in Fig. 1-C, we first extract the FOV features, textual features, and user preference features using a feature extraction model (CLIP). These features capture both the visual content of the scene and the personalized preferences derived from gaze data. The extracted features are then processed by a Stable Diffusion model, where the noise and denoising process enables the generation of content that aligns with the user's preferences.

With this approach, the DynamicEye360 adjusts the content for each FOV dynamically, continuously refining the scene based on gaze data and preference analysis. This ensures a highly personalized, immersive 360° experience that adapts as the user's attention shifts across the environment.

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

In our experiments, we use the Pico 4 RPO HMD to collect real-time gaze data from users. Our model is powered by two RTX 3090 GPUs on the server side to handle the computational demands of dynamic content generation. For network transmission, we employ the Gradio framework to enable seamless communication between the HMD and the server.

4.2 Data Analysis and Results

We compared our approach with the latest methods in 360° content generation [1, 3]. Table 1 shows that our method outperforms traditional approaches by incorporating user preferences and gaze trajectories into the content generation process. Unlike previous methods, which rely on static content, our DynamicEye360 dynamically adapts the 360° scene based on real-time gaze data, providing a more personalized and immersive experience.

Method	FID↓	LPIS↓
OmniDreamer [1] (2022)	24.5	0.41
AOG-Net [3] (2024)	18.4	0.36
Our Approach	16.7	0.32

Table 1: Comparison of performance metrics between OmniDreamer, AOG-Net, and our approach.

5 DISCUSSION

5.1 Advantages and Innovations

A key advantage of our method is its ability to dynamically adjust content based on user interaction, providing a personalized experience aligned with the user's visual focus. This dynamic adaptation significantly improves over traditional methods that rely on static content or post-session analysis, which often fail to capture evolving user preferences.

5.2 Limitations and Future Work

Despite promising results, our method relies on the accuracy of eye-tracking hardware, which may vary across users and environments. Inaccuracies could affect both preference analysis and content generation. Additionally, the computational demands of real-time processing may pose challenges when handling more complex scenes. Future work will aim to improve gaze tracking accuracy, incorporate multimodal inputs for deeper preference analysis, and enhance the capability to manage more complex content while maintaining responsiveness.

6 CONCLUSION

We proposed a personalized 360° content generation method driven by gaze tracking and user preference analysis. By adjusting content based on gaze data, our approach offers a more engaging, immersive experience compared to traditional static methods. Experimental results demonstrate that incorporating gaze trajectories and user preferences improves content personalization and user satisfaction.

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